

A Calibrated Spectral Survey of $\mathbf{L}_U$ at the Hou–Wang–Yang Profile

No second unstable eigenvalue in the odd axisymmetric sector — technical summary
(companion to preprint v1.1)

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A from-scratch, gate-validated realization of $\mathbf{L}_U$ reproduces the Hou–Wang–Yang eigenvalue $\lambda_1 = 0.113142$ to six significant figures and exhibits no second unstable eigenvalue in the odd axisymmetric sector at a calibrated sensitivity — evidence for instability order one in that sector.

Question — Beyond your verified $\lambda_1 = 0.11314203\dots$, does $\mathbf{L}_U$ (arXiv:2509.25116, your eq. for the linearized operator) have further eigenvalues with $\text{Re } \lambda > 0$ — the *instability order*, which sets the dimension of the nonuniqueness family? We survey the odd axisymmetric sector (the sector of λ_1) in floating point, with every claim gated and the instrument’s sensitivity calibrated on λ_1 itself. Everything below is evidence-grade, not certified.

Operator realization (trust anchors) — A pointwise Python realization of $\mathbf{L}_U$ in your (β, θ) coordinates, evaluated against your published repository data (read at run time; not redistributed): profile-equation residual 6.2×10^{-9} , stored eigenpair residual 1.7×10^{-5} , archive md5-gated. Exact symmetry eigenvalues as a side result: $\mathbf{L}_U(\partial_z U) = -\frac{1}{2} \partial_z U$ and $\mathbf{L}_U(U + \xi \cdot \nabla U) = -(U + \xi \cdot \nabla U)$, so no equation symmetry generates an unstable mode.

Instrument — The gradient-deflated residual landscape

$$r(\lambda) = \min_{v \in \mathcal{V}, q} \frac{\|(\mathbf{L}_U^{\text{mom}} - \lambda)v - \nabla q\|_{L^2}}{\|v\|_{L^2}} = \sqrt{\lambda_{\min}(\tilde{A}_d - \lambda \tilde{G}^\top - \bar{\lambda} \tilde{G} + |\lambda|^2 I)},$$

the second-order-relative-spectrum / residual-pseudospectrum distance function (the quantity underlying residual-DMD spectral validation; Levitin–Shargorodsky, Trefethen–Embree, Colbrook–Townsend), here realized for the divergence-free Navier–Stokes structure: $\mathcal{V}$ is an exactly divergence-free (Stokes-stream) trial space, the scalar span absorbing ∇q replaces the Leray projection, and the pencil is exactly quadratic in λ because $\langle w, \nabla q \rangle = 0$ on our quadrature domain (verified to 4×10^{-13}). A genuine eigenvalue whose eigenfunction the span represents produces a dip; the dip’s attainable depth is governed by the law below. (Motivation: in our first survey, on the ϕ -subspaces of your repository, a flagged second value $+0.036$ passed residual, jackknife, and negative-control checks yet was unmasked by this landscape as a pseudospectral shadow of λ_1 — documented in the preprint as a cautionary case study.)

Two numerical-analysis facts (known technique, silent traps) — Neither is new in isolation — weighted quadrature for axis/pole singularities is standard since Bernardi–Dauge–Maday (1999), pressure elimination by divergence-free bases is textbook — but both are invisible at the Rayleigh–Ritz pencil level and fatal to the operator-side Gram. (i) *Open-domain exactness*: for div-free decaying w , the truncated-domain $\langle w, \nabla p \rangle$ equals the cut-edge flux exactly (numerical closure 2.3×10^{-13}); the standard equatorial standoff $\theta \leq \pi/2 - 0.04$ carries the *entire* pressure term (p is odd about the equator; 96% of the operator output), while open-endpoint Gauss–Legendre on $(0, \pi/2)^2$ kills all four fluxes analytically — the dropped term collapses $6.5 \times 10^4 \rightarrow 1.3 \times 10^{-8}$. (ii) *Operator domain*: radial factors with $R(0) \neq 0$ (including shell-Hermite ψ_n inside the turning point) give $\|\mathbf{L}_U w\|^2 \sim \int \rho^{-2} d\rho = \infty$: the pencil is fine but the operator-side Gram does not exist. The mask $m(\rho) = \rho^2/(\rho^2 + 4)$ on all radial factors (admissible for any R in the Stokes-stream construction) cures it.

Trial space designed from the measured target — On the open quarter domain the sector’s Legendre families are orthogonal by parity, so v_1 ’s angular spectrum is an exact Parseval decomposition: the tail crosses 3×10^{-4} at $k = 102$. Per- k least squares under the divergence-free tie show a dual-width radial requirement ($\sigma = 2.5 \oplus 1.2$ shell-Hermite; single-width floor $8.4 \times 10^{-4} \rightarrow 7 \times 10^{-5}$), and license a taper (toroidal: 30 radial functions; poloidal $k \geq 40$: 46). Result: $N = 5366$, v_1 fit error 2.557×10^{-4} , within a 4 GB/4-core budget.

Gate ladder (selection) —

separable/heterogeneous assembly vs. direct operator columns	$\leq 5 \times 10^{-15}$	T4/T5/T6
Ritz recovery of λ_1 (two regularization cuts)	$ d = 3.0 \times 10^{-4}$	X1
$U \equiv 0$ control: $\max \operatorname{Re} \leq -\frac{1}{4}$	-0.61	X3
pressure-deflation floor on the exact eigenpair	7.6×10^{-5}	calibration
deflated operator Gram bottom, cut-stability	7 digits	PSD

Resolving power and calibration — The landscape floor obeys $r(\lambda_1) \lesssim \varepsilon_{\text{pt}} + \mathcal{A} \delta$ with δ the span’s best-approximation error of the eigenfunction and $\mathcal{A}$ the operator amplification of the missing content. Measured: $\delta = 4.4 \times 10^{-3} \Rightarrow$ flat landscape at 0.066 (no valley); $\delta = 2.6 \times 10^{-4} \Rightarrow$ valley of depth 9.3×10^{-3} . The known λ_1 thus calibrates both the attainable depth and the location accuracy for any candidate.

Results — Real axis $[-0.10, 0.30]$: a single smooth basin; fine-scan vertex at $\lambda = 0.113142$ (λ_1 to six significant figures); depth $r = 9.3141 \times 10^{-3}$, reproduced *pointwise* by an independent path (explicit arg-min field, one direct operator application, grid-level deflation: 9.3141×10^{-3}); raw values stable to 1.5×10^{-4} relative across cuts, so $\gtrsim 1\%$ local dips are detectable. Complex window $\operatorname{Re} \lambda \in [0, 0.35]$, $\operatorname{Im} \lambda \in [0.05, 0.60]$: r strictly monotone in $\operatorname{Im} \lambda$ on every grid row; minimum 1.24×10^{-2} adjacent to the real-axis valley; no off-axis local minimum. *Conclusion (scoped): no second unstable eigenvalue of $\mathbf{L}_U$ in the odd axisymmetric sector at the instrument’s stated sensitivity; instability order one in this sector.* Untouched: the even sector (surveyed only with the weaker ϕ -span), all $m \geq 1$ sectors, and rigorous certification.

What is and is not new here — The instrument is the residual-pseudospectrum / ResDMD validation principle applied to this operator, not a new method; the two facts above specialize standard axisymmetric-spectral practice. The elements we claim as new are the *resolving-power law with its calibrate-on- λ_1 protocol* (turning a null survey into a quantified-sensitivity statement), the first spectral reconnaissance of $\mathbf{L}_U$ beyond λ_1 , and the documented false-positive case study.

What we would value your view on — (1) Whether this survey and its negative are useful to your program and the community, and any errors you see, particularly in our reading of your data conventions. (2) Whether your certified finite-rank machinery could turn the sector-scoped negative into a rigorous enclosure. (3) Whether the data usage and attribution are as you would wish (your repository carries no license file; we redistribute none of it). (4) If you judge the work worth posting, we would be grateful for assistance with arXiv submission (endorsement for math.AP).

Pointers — Preprint v1.1 (12 pp.) and the full pipeline — gates, forensics, reproduction ladder with expected numbers — at <https://github.com/psalmond/hwy-spectral-survey>.